Fiscal Year:	FY 2013	**Task Last Updated:**	FY 11/01/2012
PI Name:	Long, Marshall Ph.D.		
Project Title:	Coflow Laminar Diffusion Flame		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	01/01/2012	**End Date:**	12/31/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: change in Period of Performance to 1/1/2012-12/31/2014 per NSSC information (previous POP was 10/1/11-9/30/14)--Ed., 12/30/2011		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smooke, Mitchell Ph.D. (Yale University)		
Grant/Contract No.:	NNX11AP43A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Editor's Note: Successor agreement to Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment (NNX07AB33A), per D. Stocker, NASA Glenn Research Center. Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum – namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform to gain a better understanding of the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth and oxidation processes in soot formation models. The overall goals of the research are to improve our understanding of the physical and chemical processes controlling diffusion (i.e., non-premixed) flame structure and lifting phenomena (i.e., stabilization) and to provide for rigorous testing of numerical models, including thermal radiation, soot formation, and detailed chemical kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation and radiative transfer take place.
Task Progress:	SLICE Experiments were carried out on board the ISS early in 2012. A comprehensive data set with a variety of test conditions, including different hydrocarbon fuels with varying dilution levels, different fuel nozzle sizes, and various flow conditions was obtained. A fully characterized color digital camera was used to image line-of-sight flame emissions to obtain flame properties such as the flame shape and size, lift-off height, as well as soot temperature and volume fraction. The data analysis that has been done includes quantitative determination of soot temperature and volume fraction for all sooty flames and lift-off height measurements for lifted flames. The data analysis is ongoing. The major task that remains to be done is performing similar tests using a copy of the SLICE hardware under normal gravity, which will enable direct comparisons of flame properties between 0 g flames and 1 g flames. Numerical simulation of the flames will also be performed and compared with experimental results.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Takahashi F, Kulakhmetov R, Stocker DP, Ma B, Long MB. "Microgravity Enhances the Stability of Gas-Jet Diffusion Flames." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 28 – December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 28 – December 2, 2012. https://asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Nov-2012
Abstracts for Journals and Proceedings	Ma B, Giassi D, Stocker DP, Takahashi F, Long MB. "Structure and liftoff in combustion experiment (SLICE) on the ISS" 34th International Symposium on Combustion, Warsaw, Poland, July 29-August 3, 2012. 34th International Symposium on Combustion, Warsaw, Poland, July 29-August 3, 2012. Poster session. , Aug-2012
Abstracts for Journals and Proceedings	Ma B, Long MB. "Absolute light calibration using S-type thermocouples." 34th International Symposium on Combustion, Warsaw, Poland, July 29-August 3, 2012. 34th International Symposium on Combustion, Warsaw, Poland, July 29-August 3, 2012. Presentation. , Aug-2012
Abstracts for Journals and Proceedings	Ma B, Long MB. "Novel soot volume fraction measurement through ratio-pyrometry and absolute light calibration." Presented at 5th International Workshop on Laser-induced Incandescence, Le Touquet, France, May 8-11, 2012. Proceedings 5th International Workshop on Laser-induced Incandescence, Le Touquet, France, May 8-11, 2012. http://ceur-ws.org/Vol-865/Ma.pdf , May-2012
Articles in Peer-reviewed Journals	Ma B, Long MB. "Absolute light calibration using S-type thermocouples." Proceedings of the Combustion Institute. In Press, Corrected Proof, Available online 14 June 2012. http://dx.doi.org/10.1016/j.proci.2012.05.030 , Jun-2012